Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Python · Secure Systems · AI/ML Integration · CI/CD · OOP · Production Reliability · Code Review · Compliance · Mentoring · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years delivering secure, scalable production systems in complex, compliance-critical environments - with a track record of integrating AI and machine learning to improve operational outcomes. At Entrata, led delivery of 20+ production modules across 120+ Agile sprints: Python and JavaScript OOP-based backends, CI/CD-driven quality, and AI-driven observability tooling reducing triage time 40% at 99.5% uptime for 100K+ monthly active users. Enforced 100% PHI compliance with Libsodium-encrypted pipelines and zero security incidents across 7 years. Conducted 300+ code reviews and mentored 5+ engineers. Built and shipped two AI systems from scratch: AgentProbe (open-source LLM evaluation platform with CI-integrated regression gating and performance monitoring) and LeaseI (8-agent agentic AI system with RAG pipelines, HITL escalation, and 93.6% precision across 1,240 test cases). Consistently translates complex business requirements and compliance standards into measurable, maintainable technical outcomes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python (8yr), JavaScript, TypeScript, SQL, PHP, Bash, OOP (object-oriented programming), FastAPI, Flask, REST APIs, Docker, Microservices, CI/CD (GitHub Actions, automated testing), AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Linux, Git

Security & Compliance: Secure System Design, PHI Compliance, Libsodium Encryption, Data Integrity, Access Control, Compliance-Critical Pipeline Engineering, Documentation-Based Coding Practices, Code Review Standards, Incident Management, Observability

AI & Generative AI Integration: LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Evaluation, LLM-as-Judge, Hallucination Detection, HITL, Guardrails, GenAI Tools, Prompt Engineering, ChromaDB, OpenAI API, Anthropic API, Claude Code, BERT, spaCy, NLP, PyTorch, scikit-learn, NumPy, Pandas, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Integrated AI and GenAI tools into a production agentic system (LangGraph, Python) - iterative development from requirements through CI/CD deployment with automated testing gates enforcing quality at every release.
- Translated complex operational requirements into scalable, maintainable software with deterministic guardrails; leveraged generative AI tools to accelerate development velocity and improve solution quality.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built scalable ML pipelines (Python, scikit-learn, PyTorch) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end iterative development from data ingestion through model evaluation across 190+ countries.
- Mentored 600+ students in Python, OOP, SQL, and software engineering best practices; designed monitoring dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Delivered scoped features through collaborative engineering teams** - 20+ scalable production modules across 120+ Agile sprints; OOP-based Python and JavaScript architecture designed for scalability, maintainability, and security; translated complex compliance and business requirements into measurable technical outcomes; 99.5% uptime SLA at 100K+ monthly active users.
- **Secure, compliant system design and data integrity** - 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents across 7 years); AWS S3, EC2, RDS at enterprise scale; RabbitMQ event-driven async processing with retry and idempotency guarantees; Docker containerization (50% scalability improvement); 200+ SQL query optimizations (30% performance gain).
- **AI/ML integration and GenAI tools for operational improvement** - AI-driven observability and monitoring tooling reducing triage time 40%; ML-powered data validation across compliance-critical workflows; leveraged generative AI development tools to enhance software quality and delivery speed.
- **Code review, debugging, and engineering excellence** - 300+ peer code reviews enforcing documentation-based coding practices and engineering standards; maintained sub-10 bug count for 6+ consecutive months; mentored 5+ engineers on OOP, secure coding, and continuous improvement; CI/CD via GitHub Actions with automated testing.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) with OOP design patterns; data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: GenAI Evaluation & Performance Monitoring Platform

GenAI · CI/CD · Monitoring

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · Anthropic · OpenAI | github.com/venkatalakshmiparimala/AgentProbe

Scalable GenAI integration platform: 15-endpoint FastAPI service with OOP architecture, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated automated testing gate, multi-provider LLM routing. Leverages GenAI tools throughout - zero regressions shipped across 1,240+ test cases.

Leasel: 8-Agent Agentic AI & Decision Automation System

AI/ML · Agentic · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production AI system integrating machine learning components to automate compliance decision workflows: 8-agent LangGraph pipeline with deterministic guardrails, HITL escalation for edge cases requiring human review, and CI quality gate. 93.6% precision across 1,240 compliance test cases - measurably improved decision throughput and reduced manual review workload.

AI on FHIR: NLP-Driven Enterprise Query Automation

AI/ML · NLP · Full-Stack

Python · Flask · React · Docker · BERT · spaCy · LLMs · FHIR APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI/ML integration translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - reduced retrieval time ~60%; HIPAA-compliant RBAC with secure access control and full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017